

Full Length Article

Dynamic defense model based on bionic mechanism

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ABSTRACT

Information systems have become foundational to modern critical infrastructures, where security is of paramount importance. However, with the increasing complexity of highly dynamic and large-scale environments, traditional perimeter-based defense mechanisms, which rely on external protection and reactive responses, are no longer sufficient to counter sophisticated and evolving threats. To address this challenge, this paper proposes a dynamic Defense Model based on Bionic Mechanism (DMBM), inspired by the human immune system's capability to distinguish "self" from "non-self" through adaptive and distributed mechanisms. The proposed model introduces an endogenous and active defense paradigm, integrating trust evaluation, dynamic defense, self-healing and immune memory. A dynamic threshold mechanism is further introduced to enhance adaptability to runtime variations and unknown attacks. Experimental results demonstrate that DMBM achieves True Positive Rate (TPR) of 97.45% with False Positive Rate (FPR) as low as 2.81%. Compared with NeuroShield, Active Defense Model based on Situational Awareness and Firewalls (SAF-ADM), Immune-Inspired Malware Detection Algorithm (IIMDA) and Federated, Trust-Aware, and Explainable Self-Healing (FTASH) architecture, DMBM improves detection accuracy by approximately 3%–10% while reducing the false positive rate by about 2%–3%. For unknown attacks, DMBM achieves detection accuracy improvement of approximately 2%–15%.

1. Introduction

The research on new generation security defense technologies has attracted significant global attention, with major technological powers incorporating it into national strategic development plans. As information systems grow increasingly complex and become more widely applied, the security protection capabilities of these systems face unprecedented challenges. In recent years, concerns regarding information system security have continued to rise, particularly in fields such as defense system, trusted computing, and autonomous computing. Research efforts both domestically and internationally have intensified, accompanied by continuously evolving research focuses.

However, over the past few years, the vast majority of trust evaluation and defense models proposed for information system security have been passive defense models. This traditional perimeter-based defense is a security model based on boundaries, static configuration, and heavily dependent on internal default trust assumptions, which can not fundamentally resist external attacks. The critical challenge in current information system security therefore lies in pivoting to an internal focus,

overcoming the bottlenecks of traditional perimeter defense to build a holistic security evaluation and dynamic defense system that functions from the inside out.

Shen (Wu, 2024) identified three primordial deficiencies that contribute to cybersecurity vulnerabilities and elevate its risk profile: the absence of an offensive-and-defensive security paradigm within Turing's computational theory, the lack of intrinsic security components in the von Neumann architecture, and the deficiency of security governance frameworks for critical engineering applications. To address these problems, he posited that it is imperative to consolidate the security perimeter of critical infrastructure and architect a dynamic defense system to secure strategic dominance in core technology innovation.

Therefore, guided by this philosophy and inspired by the human immune mechanism, (Wu, 2020) pointed out that the risks exposed by the extensive integration of artificial intelligence into information systems are predominantly endogenous security issues. These issues originate from inherent flaws in computer architecture, which manifest in practical applications as endogenous security vulnerabilities within the

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program control framework. The defensive capabilities of the human body are innate and deeply integrated with its physiological structure, representing an internally generated security mechanism. This bears a strong resemblance to information system security defense in terms of objectives, structure, and operational models.

As shown in Fig. 1, the human body consists of the nervous system, the immune system and the circulatory system working in coordination. Among these, the immune system performs the defense functions of the organism by recognizing and eliminating foreign threats such as pathogens and abnormal cells, establishing immune memory to maintain internal homeostasis. The nervous system is responsible for regulation and conduction within the body, transmitting electrical signals through neuronal networks to achieve sensory perception, information integration, and command issuance, thereby coordinating the functional activities of various systems. The circulatory system undertakes the transport functions of the organism, delivering oxygen, nutrients, and metabolic wastes through blood flow, regulating temperature balance, and providing essential material exchange for the survival of all tissue cells.

Thus, through its immune, nervous, and circulatory subsystems, the human body constitutes a highly efficient natural perception, defense, and self-healing system (Myakala et al., 2025). This system achieves a closed-loop process encompassing threat monitoring, defense response and resource replenishment. The synergistic mechanisms and homeostatic principles therein provide inspiration and model references for constructing dynamic information system security architectures.

Inspired by the operational mechanisms of the human body, the immune system handles defense and immune memory, the nervous system undertakes perception and real-time monitoring, and the circulatory system manages recovery strategies and resource scheduling. Together, these three architectures form a bionic-inspired defense system.

The immune, nervous, and circulatory systems collectively exhibit a tightly coupled process of perception, assessment, defense, and recovery. To translate these biological principles into a computational paradigm for information system security, we abstracted their functionalities into a unified closed-loop mechanism consisting of parameter perception, defense assessment, damage quantification, and adaptive self-healing. In this paper, we proposed a dynamic Defense Model based on Bionic Mechanism (DMBM) following the above mechanisms:

1. Conduct biomimetic signal processing of internal and external perceptual parameters, normalizing them into a fixed-format encoding for model input.
2. Based on neuronal connectivity, an associate-isolated evaluation paradigm is constructed, where interconnected entities are assessed via correlation variations and intrinsic feature deviations to identify known and unknown attacks. The mechanism enables endogenous, self-driven detection with minimal reliance on external mechanisms.

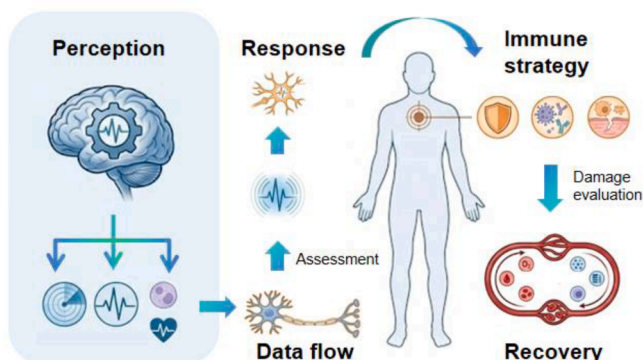


Fig. 1. The coordination of human body systems.

3. After the occurrence of aggressive behavior, the damage to the objects within the system is assessed, and a recovery strategy is formulated based on Markov Decision Process (MDP).

This paper is divided into five parts. The first section introduces the shortcomings of the traditional defense model for information systems and proposes bionic-inspired dynamic defense architecture. The second discusses current research of endogenous security and security defense models, and analyzes the shortcomings of the existing models. The third proposes DMBM, which is built upon the associate-isolated paradigm and integrates trust assessment, detection, defense, self-healing and immune memory into complete dynamic defense chain. The fourth conducts experiments focusing on detection accuracy and recovery capability to comprehensively evaluate the performance of DMBM. Furthermore, comparative analysis with NeuroShield, SAF-ADM, IIMDA and FTASH is carried out to further demonstrate the feasibility and effectiveness of DMBM. The fifth section presents a summary of our work and discusses potential future prospects.

2. Related works

Current research on bionic-inspired models predominantly centers on the integration of endogenous security mechanisms into defense architectures. The core principle of endogenous security is to embed security as an intrinsic property of the system, rather than as an externally imposed layer, typically realized through the Dynamic, Heterogeneous, and Redundant (DHR). By leveraging internal dynamism, heterogeneity, and redundancy, such systems can achieve adaptive and spontaneous defense against unknown threats, ensuring secure and reliable operation.

Along this line, existing studies have explored endogenous security from multiple perspectives. Lin et al. (2025) extended the concept to wireless environments by proposing Wireless Endogenous Security (WES) for Space-Air-Ground Integrated Networks (SAGINs), where dynamic and heterogeneous channel characteristics are exploited to enhance secure communication while balancing security and accessibility. Luo et al. (2025) further advanced the paradigm by introducing a cybersecurity resilience capacity model, which formulates endogenous security capabilities as a dynamic and quantifiable process across prevention, resistance, recovery, and evolution under varying threat conditions. Focusing on data protection, (Zhang et al., 2025) incorporated endogenous security into a mimic defense framework and proposed a dynamic encryption model, where heterogeneous algorithms and adaptive selection strategies improve uncertainty and resilience against targeted attacks.

In addition, bionic-inspired approaches have been introduced to further strengthen endogenous security mechanisms. Weng et al. (2014) developed an immune network-based cluster intelligence algorithm, enabling distributed and adaptive decision-making through local interactions, thereby enhancing scalability and robustness in dynamic environments. Similarly, (Li et al., 2024) proposed an endogenous security framework (ESUAV-NI) for Unmanned Aerial Vehicles (UAV) systems, which integrates neural-inspired rapid response with immune-inspired adaptive defense, and further incorporates an AI-driven vaccine mechanism for unknown threat prediction.

These studies collectively demonstrate that endogenous security provides a fundamental framework for intrinsic protection through structural and functional design. However, the increasing complexity and intelligence of cyber threats call for more advanced adaptive defense mechanisms, motivating the introduction of bionic-inspired paradigms such as artificial immune systems.

The artificial immune system (AIS) is an intelligent computing methodology inspired by the biological immune system and belongs to an important branch of bionic-inspired computing. Tao et al. (2025) proposed NeuroShield, an dynamic defense model inspired by neural control mechanisms, which integrates fine-grained perception, adaptive

learning, and real-time threat mitigation to proactively defend web applications. But NeuroShield lacks self-healing mechanism and does not evaluate system recovery capabilities, limiting its resilience in production environments where rapid restoration from attacks is essential.

Saurabh and Verma (2016) proposed an Efficient Proactive Artificial Immune System based Anomaly Detection and Prevention System (EPAADPS), aiming to identify and prevent novel and unknown anomalies. The system improves the limitations of traditional negative selection algorithms in covering self/non-self space by introducing self-tuning mechanisms and detector power. Meanwhile, it adopts multi-agent collaboration and voting mechanisms to effectively reduce the false alarm rate and possesses dynamic defense capabilities against unknown attacks. However, in practical deployment, especially when facing large-scale network traffic, its training efficiency and scalability still lack strong support.

Vidal et al. (2018) proposed a method for mitigating Denial of Service (DoS) flooding attacks using AIS. The approach builds a distributed sensor network tailored to the requirements of the monitored environment, simulating innate and adaptive immune responses in humans, the establishment of quarantine areas, and the construction of immune memory. The system employs entropy-based detection methods and Holt-Winters forecasting techniques to identify anomalies, and utilizes a multi-agent architecture composed of H-detectors and A-detectors to collaboratively detect and mitigate threats. However, when facing constant-rate flooding attacks, the entropy tends to stabilize, causing the detection mechanism to fail.

The (Li et al., 2023) proposed a dynamic defense model which integrates situational awareness with firewall. It analyzes security logs to assess system states and identifies potential attackers through alarm accumulation within a time window using predefined thresholds. Once triggered, suspicious IP addresses are automatically blocked, forming a closed-loop defense process. However, the model heavily relies on threshold settings and statistical aggregation, making it sensitive to parameter configuration. It often exhibits delayed response in early stages and quickly reaches performance saturation after threshold triggering.

Prajwalasimha et al. (2025) proposed FTASH, which combines federated learning, Bayesian trust calibration, and explainable reinforcement learning for self-healing in cyber-physical systems. It offers privacy-preserving distributed training, trust-aware aggregation against malicious nodes, and interpretable control policies. Nevertheless, FTASH incurs high communication overhead and recovery delays from federated consensus, consumes substantial computational resources due to Explainable Artificial Intelligence (XAI) and uncertainty estimation, and suffers from detection precision degradation under differential privacy noise. It also lacks a local autonomous recovery mechanism and remains vulnerable to coordinated attacks that violate its honest-majority trust assumption.

Zhou et al. (2025) proposed a resource efficient Moving Target Defense (MTD) for low rate Distributed Denial of Service (DDoS) attacks in edge clouds using deep reinforcement learning. Its strengths include three lightweight MTD mechanisms built on container functionalities and a Deep Q-Network (DQN) based Markov decision process to balance security and resource use. Experiments show clear improvements in security and service quality with low overhead. However, the approach cannot withstand attackers with vastly greater resources, may fail against insiders who know the MTD mechanisms, and assumes stable network environments and user behaviors. It also relies on time step based decisions rather than event triggered responses, which can cause delays in highly dynamic scenarios.

Rufus et al. (2016) proposed an autonomic computing system that integrates a rule based policy engine with artificial immune systems to achieve self configuration, self healing, self optimization, and self protection. The strengths include using AIS for anomaly detection to distinguish normal from malicious behavior, employing a context monitor to track system state, and leveraging the Jess rule engine for

policy enforcement. This architecture reduces the need for human intervention in complex networked systems. However, the approach has several limitations. It remains largely conceptual with no experimental validation or performance evaluation. The reliance on a rule based policy engine may introduce rigidity when facing novel or evolving threats. AIS based detection can generate false positives and incurs computational overhead. The system also assumes that policies can be predefined for all situations, which is difficult in dynamic environments. Furthermore, the architecture does not address scalability or deployment challenges in real world networks.

Adhikari et al. (2025) proposed an immune-inspired malware detection algorithm (IIMDA) for edge-based industrial Internet of Things (IIoT) cybersecurity. Its strengths include an antigen-antibody matching mechanism and a dynamic memory cell repository that enable adaptive malware detection and low-latency threat response. Experimental results on the Bot-IIoT dataset demonstrate high detection accuracy, low false alarm rates, and efficient operation in resource-constrained edge environments. However, the evaluation is limited to a single benchmark dataset and does not provide quantitative validation against truly unknown attacks.

These bionic-inspired models provide the architectural foundation for our research. In summary, the existing models still have the following shortcomings:

1. Existing models mostly rely on external feature extraction or preset thresholds for detection, failing to establish a preprocessing mechanism for internal and external perception.
2. The defense mechanisms are largely dependent on externally driven strategy for defense assessment and response. This external reliance prevents the formation of intrinsic and dynamic defense, making it difficult to achieve decision-making and adaptive evolution in dynamic environments.
3. Those methods generally focus on attack detection and mitigation, without assessing the internal damage caused by attacks or incorporating efficient recovery strategies. This results in a significant shortfall in system resilience. The following section presents the detailed design of DMBM, which addresses these limitations through its core mechanisms.

3. Biomimetic security methodology

To provide interpretation of the biological inspiration underlying DMBM, a functional mapping between biological defense mechanisms and DMBM components is established, as summarized in Table 1.

DMBM adopts their functional principles to organize perception, assessment, defense, and recovery within information systems. In order to complete the improvements above, we divided DMBM into three modules according to the human body functions. Fig. 2 shows the framework of the model. The process of each module is as follows:

Table 1
Mapping Between Biological Functions and DMBM.

Biological Principle	Immune Function	DMBM Realization
Antigen	External stimuli	Parameter encoding
Antibody	Recognize abnormal patterns	Defense strategy
Immune tolerance	Reduce false responses	Dynamic threshold mechanism
Synaptic plasticity	Adaptive learning	STDP feedback
Immune response	Defense action selection	Defense strategy response
Self-healing	Damage recovery	MDP-based recovery policy
Immune memory	Retain historical knowledge	Characteristics recognition and optimal policy replay

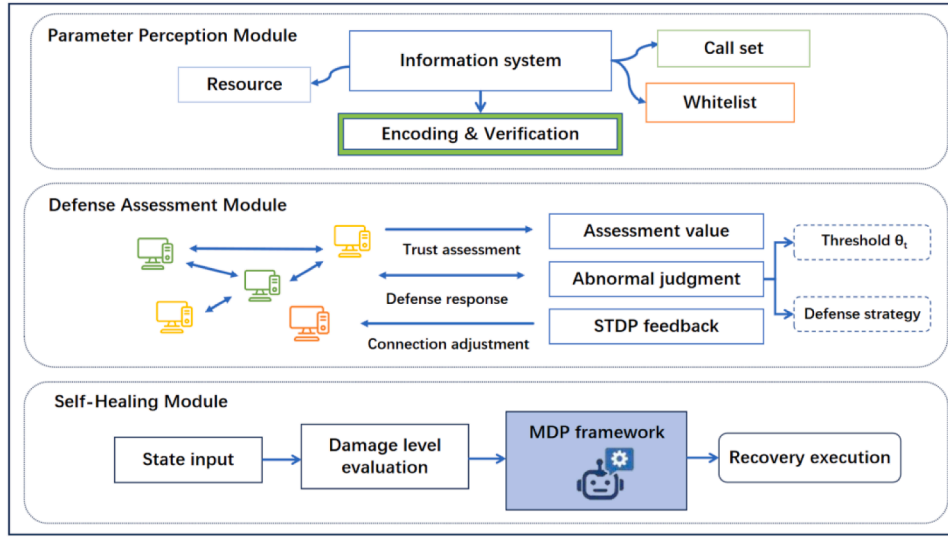


Fig. 2. DMBM architecture.

1. **Parameter Perception Module (PPM)**, based on the perception mode of the human nervous system towards the environment, multimodal parameter perception module for information system has been constructed. The module categorizes parameters into internal and external perception parameters, and uses built-in receptors to collect and preprocess parameters.
2. **Defense Assessment Module (DAM)**, after parameter preprocessing, the human body performs self and non-self identification and conducts assessment. In this module, inspired by the Spike-Timing-Dependent Plasticity (STDP) (Rahman and Yusoff, 2024) of neurons, an associate-isolated bionic architecture performs trust assessment on objects in system and makes corresponding countermeasures according to the defense strategy. In addition, an immune memory mechanism is incorporated to capture and retain characteristics of unknown attacks.
3. **Self-Healing Module (SHM)**, mimicking the human immune system, when an intrusion signal is detected, the system isolates the affected objects to prevent the threat from spreading, and the self-healing module is activated. This module performs damage assessment to determine the extent of the damage, and data. Based on the extent of damage, this module dynamically allocates backup resources through Markov decision process framework to reconstruct the damaged components.

The parameter perception module collects and preprocess system parameters for trust assessment. The defense assessment module conducts trust assessment, executing responses and immune memory based on defense strategy. The self-healing module evaluates the extent of damage and dynamically reconstructs components. These three modules form an integrated detection, defense and recovery framework.

3.1. Parameter perception module

During environmental perception, the human nervous system receives physical or chemical stimuli from both internal and external environments through receptors distributed throughout the body and converts them into neural electrical signals. These signals are transmitted via afferent neural pathways (such as the spinal cord and brainstem) to primary processing centers in the brain, where neurons in the cerebral cortex perform multi-level integration and combine them with memory and experience to form perception (Berntson and Khalsa, 2021).

Similarly, in the information system, various functional application

software collectively constructs a complete operational system through interaction with the internal and external environment. Throughout this process, the system continuously generates and dynamically records parameters that reflect its operational status, behavioral patterns, and interaction characteristics. In this paper, these parameters are categorized into two types: kernel-mode and user-mode parameters. Kernel-mode parameters refer to the global parameters, such as system calls, and privileged instruction invocation. User-mode parameters refer to local parameters, such as CPU, I/O, memory usage and function calls.

After parameter acquisition, the data are normalized and encoded into a unified representation, mimicking the biological perception mechanism of transforming stimuli into signals.

As shown in Fig. 3 the parameter encoding is divided into three segments, namely resource occupation, call set and verification code.

Firstly, built-in receptors are injected into each object, and the original parameter set of the i th object in the t -th time is represented as:

$$P_i(t) = [C_t, M_t, I_t, F_t, S_t, P_t] \quad (1)$$

C_t , M_t and I_t represent the CPU, memory usage and I/O. These three parameters comprehensively reflect the load occupied in the system. F_t denotes the set of the function call, S_t is the system call sequence. P_t is the capability. The call set reflects the event activities within the system.

In summary, the parameter encoding of the i th object at the t -th time is expressed as:

$$E_i(t) = \text{BLAKE3}(\text{Encode}(C_t, M_t, I_t, F_t, S_t, P_t)) \parallel V_t \quad (2)$$

To enhance representation efficiency and ensure robust feature encoding, $E_i(t)$ employs BLAKE3 as a unified outer hashing function to generate a fixed-length representation of runtime parameters. All runtime parameters are serialized using a length-prefixed encoding, where each field is represented as "length || value", ensuring a unique representation. In addition, V_t denotes verification code derived from consistency checking between whitelists and runtime behaviors. This encoding integrates parameters from the information system, and its integration logic mimics the biological neural system of converting sensory information into unified electrical signals.

3.2. Defense assessment module

After the parameter encoding is generated, the defense assessment module simulates the feedforward inhibition and gain modulation mechanisms of neural synapses. This enables real-time assessment and modulation of the encoding. By judging encoding characteristics based

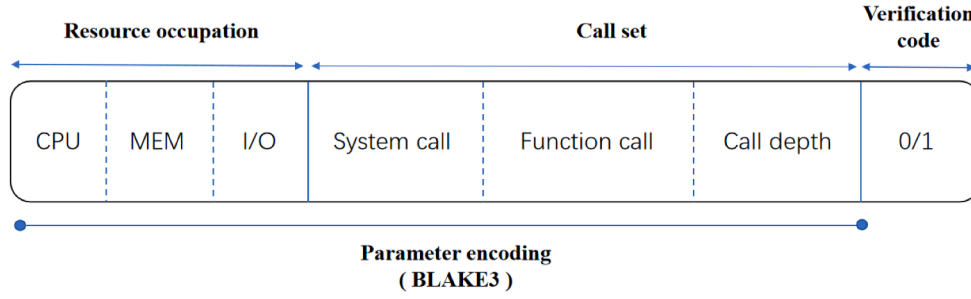


Fig. 3. Parameter encoding composition.

on synaptic plasticity rules (Song et al., 2000), the module performs self and non-self recognition. If abnormal or threatening patterns are identified, it immediately triggers a dynamic defense response, thereby achieving protection of system security.

3.2.1. Trust assessment

According to the “overall connectivity to disconnection” mechanism (initial exuberant synaptogenesis followed by activity-dependent pruning), this module implements a dynamic associate-isolated paradigm for defense assessment.

As shown in Fig. 4, during the initialization stage, all monitored objects in the system are associated with each other by external receptors. In the assessment stage, each receptor generates parameter encoding, and calculates the auto-correlation according to the context. First, the Hamming distance is used to transform the encoding sequence, which is denoted as follows:

$$H_i(t) = \frac{1}{n} \sum_{k=1}^n \text{popcount}(E_t[k] \oplus E_{sl}[k]) \quad (3)$$

where n represents the length of encoding, $E_t[k]$ is the k -th bit in encoding, E_{sl} is the baseline hash value with the same length as E_t . Then the auto-correlation of i th object at t -th time can be expressed as:

$$a_i(t) = \sum_{t=1}^{N-l} (H_i(t) - \bar{H})(H_i(t-l) - \bar{H}) / \sum_{t=1}^N (H_i(t) - \bar{H})^2 \quad (4)$$

where N is the length of the time series, l represents the time lag, $\bar{H}$ is the mean of Hamming distance in the time series. The normalization of $a_i(t)$ is carried out via the following formula:

$$A_i(t) = \begin{cases} 1.0 & t = 1 \\ 0.5 + 0.5 \cdot \text{sign}(a_i(t)) \cdot \sqrt{|a_i(t)|} & t \geq 2 \end{cases} \quad (5)$$

As part of the auto-correlation computation, the encoding is transmitted by the built-in receptors to the external receptors shared between objects, where the cross-correlation is performed. Similarly, the Hamming distance similarity between i th and j -th objects is calculated, which is defined as follows:

$$HS(i, j) = \frac{1}{N} \sum_{t=1}^N \left(1 - \frac{d_{\text{Hamming}}(E_i(t), E_j(t))}{n} \right) \quad (6)$$

Based on the Hamming distance, the change pattern of i th object in time series is defined as follows:

$$\Delta_i = [d_{\text{Hamming}}(E_i(1), E_i(2)), \dots, d_{\text{Hamming}}(E_i(N-1), E_i(N))] \quad (7)$$

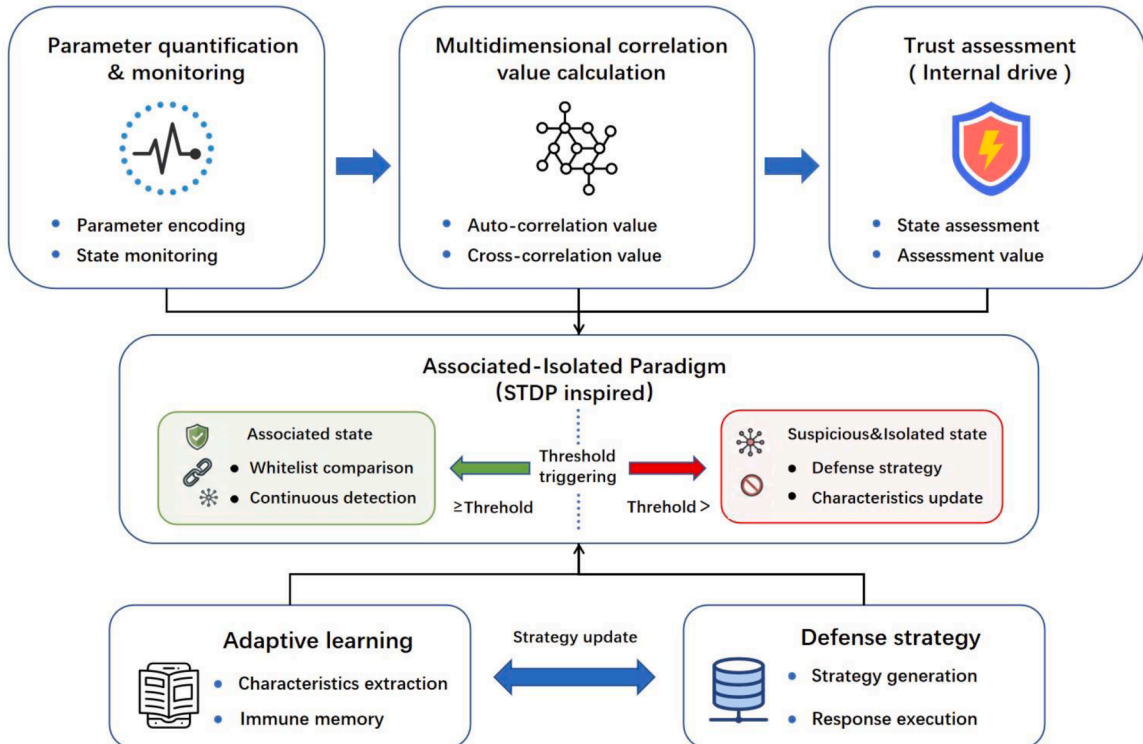


Fig. 4. Defense assessment diagram.

Therefore, the context similarity CS between i -th and j -th objects is expressed as:

$$CS(i, j) = 1 - \frac{\|\Delta_i - \Delta_j\|_1}{n \times (N - 1)} \quad (8)$$

where CS reflects whether the rhythm of state changes between two objects are synchronized, which can reveal whether they are driven by the same event or share the same resource bottleneck. To measure the similarity in the degree of change between two objects, the entropy is calculated based on the Hamming distance to illustrate the stability. The sequence entropy of the i -th object at t -th time is calculated from Δ_i as follows:

$$e_i(t) = - \sum_{t=1}^N \frac{d_{Hamming} \times n(d_{Hamming})}{T} \log_2 \frac{d_{Hamming} \times n(d_{Hamming})}{T} \quad (9)$$

where $n(d_{Hamming})$ denotes the statistical number of this Hamming distance in Δ_i , and T is the sum of Hamming distances in Δ_i . Defining the maximum entropy value in the sequence as e_{max} , the degree of change of the i -th object at t -th time is expressed by the normalized entropy as:

$$D_i(t) = \frac{e_i(t)}{e_{max}} \quad (10)$$

In summary, the entropy similarity between i -th object and j -th object can be expressed as:

$$ES(i, j) = 1 - b|D_i(t) - D_j(t)| \quad (11)$$

where b is the stability factor. ES captures the degree of divergence in parameter changes among objects. A higher value indicates higher correlation between the objects. Combining HS , CS and ES , the cross-correlation value between i -th and j -th objects at t -th time is obtained as:

$$c_{ij}(t) = w_1 \cdot HS(i, j) + w_2 \cdot CS(i, j) + w_3 \cdot ES(i, j) \quad (12)$$

where the weights w satisfy $w_1 + w_2 + w_3 = 1$. The magnitude of the cross-correlation value reflects the connection strength between objects.

Following STDP in synaptic plasticity, the connection strength between neurons is adjusted by the precise time difference in the occurrence of action potentials. In this module, building upon this biological learning rule, trust assessment of object is conducted through evaluation of the auto-correlation and cross-correlation values.

Based on the above definitions, the comprehensive cross-correlation value of i -th object at t -th time is defined as:

$$c_i^{com}(t) = r_i(t) \cdot \frac{1}{M} \sum_{j \neq i}^M c_{ij}(t) \quad (13)$$

where $r_i(t)$ is the reward and punishment factor. M denotes the number of objects associated with the current one. According to the DMEM definition, during the system initialization stage, all objects are inter-related. Once it reaches the assessment stage, if the $c_i^{com}(t)$ is below the threshold, it will be disconnected. In the special case where an object is isolated, M is set to 1.

The proposed model dynamically adjusts the connection strength between objects according to their behavioral consistency. In STDP, repeatedly correlated neuronal activities strengthen synaptic connections through Long-Term Potentiation (LTP), whereas weakly correlated or inconsistent activities weaken synaptic connections through Long-Term Depression (LTD). Analogously, in the proposed framework, objects exhibiting behavior consistent with their operational context are rewarded by strengthening their connection strength, while objects showing abnormal or weak connection strength are penalized through connection suppression. Therefore, the $c_i^{com}(t)$ is adaptively regulated by reward and punishment factor $r_i(t)$, which serves as a computational analogue of the LTP/LTD learning process and is defined as follows:

$$r_i(t) = \begin{cases} 0 & \text{Isolated} \\ (1 - L_-) \cdot \exp\left(-\frac{\tau}{\tau_-}\right) \cdot EWMA_i(t) & T_i^a(t) < \theta_t \\ (1 + L_+) \cdot \exp\left(-\frac{\tau}{\tau_+}\right) \cdot EWMA_i(t) & T_i^a(t) > \theta_t \end{cases} \quad (14)$$

where θ_t represents the dynamic threshold, $T_i^a(t)$ is the assessment value of the i -th object at t -th time. At each detection time, the θ_t is compared with $T_i^a(t)$. L_+ and L_- are the maximum learning amplitudes for LTP and LTD. τ_+ and τ_- are the time constants for LTP and LTD. τ denotes the inter-event time for abnormal or normal behaviors. $EWMA$ is the Exponentially Weighted Moving Average, the formula is as follows:

$$EWMA_i(t) = \varepsilon \cdot EWMA_i(t-1) + (1 - \varepsilon) \cdot T_i^a(t) \quad (15)$$

where ε is the smoothing factor. In summary, $T_i^a(t)$ can be expressed as:

$$T_i^a(t) = A_i(t) \cdot c_i^{com}(t) \quad (16)$$

where $r_i(0)$ is set to 1, the magnitude of $T_i^a(t)$ is determined by both the instantaneous state and the historical context of the object.

In summary, when $T_i^a(t) < \theta_t$, the object is considered behaviorally inconsistent with its associated context. This condition is analogous to LTD, $r_i(t)$ decreases the $c_i^{com}(t)$, gradually reducing the connection strength between the object and its neighbors.

Conversely, when $T_i^a(t) > \theta_t$, the object exhibits behavior consistent with the normal operational context. This condition corresponds to LTP, $r_i(t)$ enhances the $c_i^{com}(t)$, strengthening the connection strength and increasing the trust degree of the object.

3.2.2. Defense response

During the assessment stage, the defense response is orchestrated in concert with neural signaling to screen for self and non-self entities, based on effector directives from the neuronal level. As shown in Table 2, to efficiently perceive threats, we have classified five types of attack categories in the information system and constructed a defense strategy.

In a manner consistent with the encoding for object parameters, attack characteristics are also encoded based on anomalous resource occupancy, target functions, anomalous system calls and capability. We defined that the defense strategy will perform characteristics recognition on objects in the following three situations:

1. At each detection round, if $T_i^a(t)$ of the i -th object falls below the dynamic threshold θ_t , the defense strategy triggers a response. The object is flagged as suspicious and undergoes characteristics comparison. If it matches none of the known attacks, continuous feature comparison is performed at each subsequent detection time until its $T_i^a(t)$ rises above the dynamic threshold.
2. When the i -th object is identified as a known attack, the defense strategy immediately isolates it and transfers it to blacklist, thereby removing it from further detection cycles.
3. If an object enters an isolated state during detection but no known attack is detected, the defense strategy performs characteristics recognition on its anomalous parameters and updates the strategy to form immune memory.

3.3. Self-healing module

3.3.1. Damage assessment

After assessment and defense response, objects require recovery based on the extent of object damage. Inspired by the cellular hierarchical damage response mechanisms (Hároníková et al., 2019), we introduced the Process Damage Index (PDI). As shown in Fig. 5, PDI quantifies the damage by assessing three metrics: collaborative

Table 2
Attack characteristics.

Attack mode	Characteristics	Target call/capability
Buffer overflow	The cover function returns the address and hijacks the program execution flow.	execve("/bin/sh"), libc.system
SQL injection	Inject malicious Structured Query Language (SQL) statements into database queries to achieve unauthorized data access or operations.	substring(),updatexml()
File inclusion	Insufficient validation of input in dynamic file inclusion enables attackers to control the file path.	include(), include_once()
Over permission	After gaining privilege access, the attacker bypasses permission checks to manipulate higher privilege content.	cap_sys_admin,execve("/usr/bin/sh")
DDoS	Flooding the target server with excessive requests to exhaust resources and disrupt services.	malloc(),mysql_query(),curl_exec()

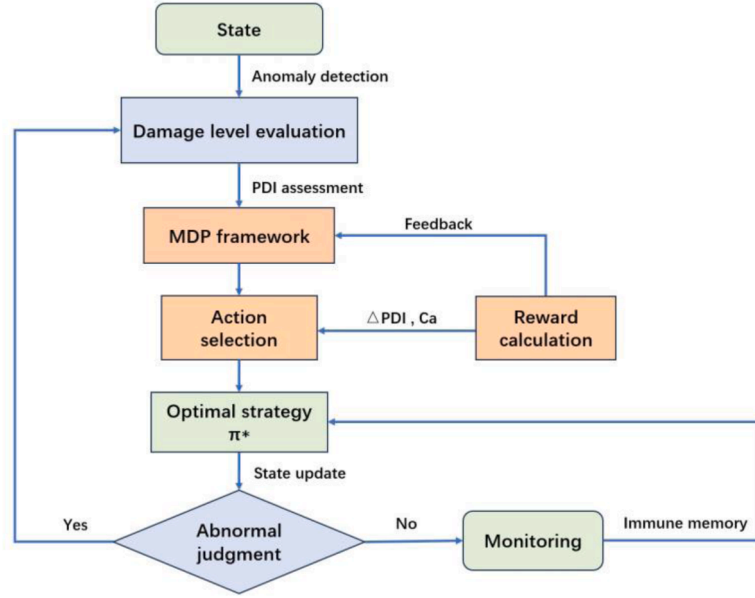


Fig. 5. Self-healing flowchart.

relationship damage (CRD), resource management impairment (RMI), and stability degradation (SD). PDI is defined as follows:

$$PDI_i = 1 - \prod_{k=1}^3 (1 - m_k S_{ik}) \quad (17)$$

where S_{ik} represents the set of three evaluation indicators for the i th evaluation object, and m_k denotes the weights of S_{ik} , which are determined by expert experience and satisfy $\sum_{k=1}^3 m_k = 1$. S_{ik} can be expressed as follows:

$$S_{ik} = \{CRD_i, RMI_i, SD_i\} \quad (18)$$

The CRD measures the degradation of collaborative interactions between objects caused by disrupted dependencies. Considering that the system interaction network is a directed graph, expressed as $G = \{V, Z\}$. Where V is the object, Z represents the dependency relationship of objects. The CRD of i th evaluation object can be defined as:

$$CRD_i = 1 - \frac{\sum_{j \in V(i)} f_{ij} Z_{ij}}{\sum_{j \in V(i)} Z_{ij}} \quad (19)$$

where $V(i)$ indicates the set of connection with the i th evaluation object, f_{ij} represents the connection strength between i th and j -th evaluation objects, and $Z_{ij} \in \{0, 1\}$ denotes whether the link is disrupted. When $\sum_{j \in V(i)} Z_{ij} = 0$ (isolated object), CRD is set to 1.

The RMI evaluates the degradation of resource management capability caused by abnormal consumption of computational resources and irregular execution behaviors. Let K denotes the number of monitored resource types, including CPU, memory, I/O resources, as well as

function/system call activities. The RMI of the i th object is defined as:

$$RMI_i = \frac{1}{K} \sum_{k=1}^K \frac{|U_{ik} - \mu_k|}{\sigma_k} \quad (20)$$

where U_{ik} is the observed utilization or activity duration of resource k for the i th object, μ_k is the expected value of the resource under normal operation, and σ_k represents the standard deviation of the resource behavior. The larger RMI indicates a greater deviation from normal resource usage patterns, reflecting a higher level of resource management impairment.

The SD characterizes the deterioration of object runtime stability caused by frequent execution errors or abnormal events, which is expressed as follow:

$$SD_i = 1 - e^{-\lambda \tau} \quad (21)$$

where λ is a sensitivity factor controlling the influence of runtime errors on stability degradation.

3.3.2. Recovery strategy

This paper proposes three tailored remediation schemes based on PDI. As shown in Table 3, we defined backup units as those storing backup data of adjacent objects and self-healing units as functional units capable of replacing compromised object to maintain system continuity.

The selection of the activation path for self-healing process plays a critical role in the efficiency of the system self-recovery process. To address the inherent rigidity and single-objective limitation of traditional distance-oriented activation path selection methods, we proposed an optimal recovery strategy based on a multi-objective MDP

Table 3
Recovery strategy.

Damage level	Strategy	PDI range
Mild damage	The compromised object conducts malicious behavior detection, followed by self-recovery using locally stored backup data.	[0,0.3]
Moderate damage	The compromised object activates its designated backup units, which restores the damaged object to a functional state by deploying the most recent backup data.	[0.3,0.7]
Severe damage	The compromised object initiates a coordinated activation of multiple self-healing units, which collectively replace the failed object to sustain system stability.	[0.7,1]

(Puterman, 1994). The MDP enables the system to dynamically select optimal recovery actions while considering the long-term recovery efficiency.

The set of five-tuple definitions for MDP is defined as (S, A, P, R, γ) , where S denotes the state space, A represents the action space, P is the state transition probability, R is the reward function, and γ is the discount factor. The state space is defined as (s_1, s_2, s_3) which is divided into three damage levels in Table 3.

The action space (a_1, a_2, a_3) corresponds to the available recovery strategies derived from the impairment levels. Each action represents a different recovery mechanism with varying recovery effectiveness and resource costs. After executing a recovery action, the system transitions from state s_t to s_{t+1} . The transition reflects the improvement or deterioration of the object condition. Formally, the transition probability is defined as $P(s_{t+1}|s_t, a_t)$. The probability is estimated from historical recovery records by counting the observed state transitions after applying a recovery action.

To evaluate the effectiveness of different recovery actions, a reward and punishment function is introduced by considering recovery improvement and recovery cost, which is defined as:

$$RP_t = \alpha \Delta PDI - \beta C_a \quad (22)$$

where ΔPDI represents the improvement of recovery action, C_a denotes the recovery cost associated with action a , α and β are weighting coefficients. The objective of the MDP framework is to determine the optimal recovery strategy. The optimal strategy π for maximizing the expected cumulative reward can be expressed as:

$$V^\pi(s_t) = E \left[\sum_{t=0}^{\infty} \gamma^t RP_t \right] \quad (23)$$

where γ is the discount factor controlling the importance of future rewards. Based on the optimal policy, the system can generate a recovery strategy path $\{a_{t_1}, a_{t_2}, \dots, a_{t_n}\}$, which enables adaptive and efficient system self-healing under different damage conditions. Based on this formulation, the optimal recovery policy is obtained by maximizing the expected return over all possible actions, defined as:

$$\pi^*(s_t) = \arg \max_{a \in A} [RP_t + \gamma V^\pi(s_{t+1})] \quad (24)$$

The optimal policy π^* is derived using the Bellman optimality principle. Specifically, for each recovery state, the expected cumulative reward associated with all candidate recovery actions is evaluated, and the action that maximizes the expected cumulative reward is selected as the optimal recovery decision. After the optimal policy is successfully executed, the module forms an immune memory. When a similar behavior is identified again by DAM, the corresponding optimal policy can be directly retrieved and applied.

4. Experiments

4.1. Experimental setup

Our experimental environment is based on 64-bit Ubuntu 22.04 system with Intel 12th Gen Core i7-12700H CPU and 16GB RAM. To ensure isolation and reproducibility, the simulation samples are constructed using Docker Desktop 4.53. Dockerfiles are employed to define application environments and dependencies, while Docker Compose is used to orchestrate multiple containers. Within this environment, a set of lightweight containerized applications is implemented to emulate typical system application software, including web services, file operations, database interactions, and network communications. The vulnerability platform Webgoat (OWASP Foundation), implemented as fully compatible with the Docker environment, is utilized as the attack testing platform, providing diverse web and information system vulnerabilities for attack simulation.

Although the Docker and WebGoat environment provides a controllable and reproducible platform for security evaluation, the experimental setup cannot fully reproduce the heterogeneity and operational complexity of real-world production systems. However, Docker preserves realistic runtime behaviors, including process execution, system calls, resource utilization, and service interactions, while WebGoat provides representative attack scenarios. Consequently, the generated attack activities exhibit behavioral patterns and execution traces that closely resemble those observed in real-world systems. Since DMBM performs trust assessment based on runtime behavioral characteristics, the current environment provides a suitable basis for evaluating the effectiveness of the proposed framework.

To evaluate the capability of DMBM against threats, both known and unknown attack scenarios are generated within the WebGoat. The known attack set consists of the attack categories listed in Table 2. Unknown attacks are constructed by modifying attack payloads, system call sequences, and resource consumption patterns of these attack categories. These variants are not incorporated into the predefined defense strategy database. Consequently, although the attack categories are known, their runtime behavioral characteristics were not previously identified by the defense assessment module. The generation strategies for each unknown attack category are summarized in Table 4.

During system operation, key runtime parameters are collected and monitored using extended Berkeley Packet Filter (eBPF) (Kim et al., 2025). By leveraging hook functions, critical system behaviors can be selectively traced, enabling fine-grained observation of system states under attack scenarios.

In addition, instead of adopting a conventional static threshold, a dynamic threshold is introduced to determine behaviors. Compared with static thresholds, dynamic thresholds better reflect the detection paradigm observed in biological immune systems. The proposed threshold incorporates three key factors: adaptive defense capability, threat level, and PDI. The definition is as follows:

$$\theta_t = \theta_0 \cdot \log_2 \left(1 + \frac{W_{defend}(t)}{W_{loss}} \right) \cdot \frac{1}{1 + PDI} \quad (25)$$

where θ_0 is the benchmark threshold, $W_{defend}(t)$ is the number of attack behaviors defended by DMBM, W_{loss} denotes the total number of attack behaviors in the system, and $\overline{PDI}$ represents the average damage index of the system. This threshold reflects the relationship between the model's dynamic defense capability and the degree of system damage.

The experimental parameters employed in this paper are summarized in Table 5. Parameters are determined through preliminary experiments and empirical tuning to balance detection sensitivity and system stability.

Table 4
Unknown Attack Generation Settings.

Attack mode	Perturbation operators	Variant characteristics
Buffer overflow	Payload length: varied from 500 to 2000 bytes (step size = 100). Return address: randomized within valid stack address space (e.g., $0 \times 7fff0000 - 0 \times 7fffffff$).	Exhaustive combination of length steps and random address offsets.
SQL injection	Keyword substitution: (e.g., UNION $\rightarrow$ UNION ALL, SELECT $\rightarrow$ DISTINCT). Logical expression variation: (e.g., $1 = 1 \rightarrow 2 = 2$, OR $\rightarrow$ AND).	Combinatorial enumeration of the syntactic rules.
File inclusion	Path depth: randomized from 1 to 5 parent directories (e.g., $.././.././$). Encoding transformation: Uniform Resource Locator (URL) encoding and Base64 transformation.	Uniform random sampling over path depths and encoding types.
Over permission	Permission combinations: derived from 3 distinct privilege roles (guest, user, admin). Execution sequences: 6 possible permutation orders of these roles (e.g., guest $\rightarrow$ user $\rightarrow$ admin, guest $\rightarrow$ admin $\rightarrow$ user).	Permutation orders combined with random role assignments.
DDoS	Request intervals: drawn from Exponential distribution (mean = 0.5 s to 5 s). Concurrent connections: 10 to 1000 (step = 50). Packet rates: 1 to 100 Mbps (step = 5 Mbps).	Stratified random sampling across low/medium/high traffic zones.

Table 5
Parameter Definition.

Parameter	Value	Description
w_1	0.4	Weight of Hamming similarity HS
w_2	0.4	Weight of context similarity CS
w_3	0.2	Weight of entropy similarity ES
L_-	0.2	Maximum depression amplitude
L_+	0.2	Maximum potentiation amplitude
τ_+	30	LTP time constant
τ_-	15	LTD time constant
ϵ	0.95	EWMA smoothing factor
m_1	0.3	Weight of CRD
m_2	0.4	Weight of RMI
m_3	0.3	Weight of SD
λ	0.95	Sensitivity factor in SD
α	0.7	Reward coefficient in MDP
β	0.3	Cost coefficient in MDP
γ	0.9	Discount factor in MDP
b	1.0	Stability factor in ES
θ_0	0.55	Benchmark threshold

4.2. Threshold test

To verify the effectiveness of the dynamic threshold, we compared the impact of the benchmark threshold θ_0 (static threshold) and the dynamic threshold θ_t on the attack detection accuracy of the model. The experiment consisted of 100 evaluation containers, among which 25% are malicious containers that exhibit the attack behaviors listed in Table 2. The detection process is conducted over 30 rounds and repeated

50 times until the detection accuracy gradually converges to a stable state.

As shown in Fig. 6, the detection accuracy under the dynamic threshold mechanism is consistently higher than that under the static threshold. In the early detection periods, both methods exhibit relatively lower detection rates due to limited behavioral observations. However, as the detection proceeds, the dynamic threshold adjusts according to the system state and gradually improves the discrimination capability of the model.

Specifically, the dynamic threshold enables the system to adaptively respond to changes in behavior and system conditions, thereby reducing the probability of misclassification caused by fixed threshold settings. As the detection rounds increase, the detection accuracy under the dynamic threshold stabilizes at a higher level and demonstrates faster convergence. This result indicates that the proposed dynamic threshold mechanism can effectively enhance the model's ability to identify malicious behaviors and maintain stable detection performance.

4.3. Sensitivity analysis

To further evaluate the robustness of DMBM under different threshold settings, a threshold sensitivity analysis is conducted using four benchmark threshold (0.35, 0.45, 0.55, and 0.65). For each threshold, the detection accuracy and FPR are measured across five attack categories in Table 2. The experimental results are shown in Table 6.

As shown in Table 6, the threshold value significantly influences the detection performance of DMBM. In general, increasing θ_0 generally

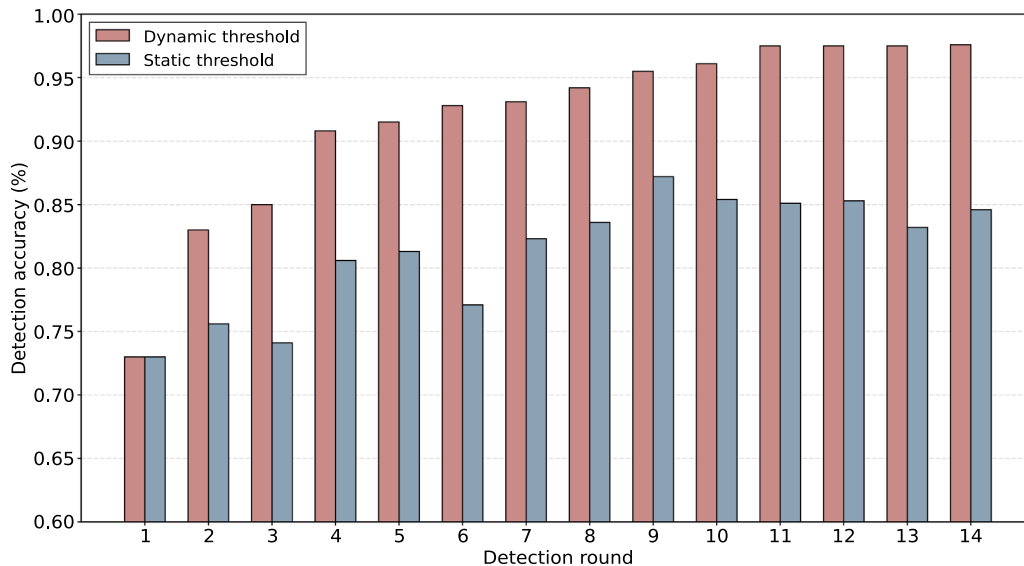


Fig. 6. Comparison of attack detection accuracy under dynamic and static thresholds.

Table 6
Sensitivity Analysis under Different Threshold Settings.

Attack Type	Metric	$\theta_0 = 0.35$	$\theta_0 = 0.45$	$\theta_0 = 0.55$ (DMBM)	$\theta_0 = 0.65$
DDoS	Accuracy (%)	95.11 ± 1.27	95.22 ± 1.13	97.83 ± 0.69	98.01 ± 0.78
	FPR (%)	0.82 ± 0.10	1.09 ± 0.11	1.27 ± 0.06	1.72 ± 0.11
SQL injection	Accuracy (%)	90.96 ± 1.51	93.21 ± 1.26	96.82 ± 0.87	93.89 ± 1.29
	FPR (%)	0.93 ± 0.13	0.96 ± 0.08	1.37 ± 0.09	1.86 ± 0.15
Buffer overflow	Accuracy (%)	87.17 ± 1.69	90.93 ± 1.51	95.77 ± 0.92	96.25 ± 0.81
	FPR (%)	0.34 ± 0.04	0.72 ± 0.07	1.13 ± 0.13	1.44 ± 0.11
File inclusion	Accuracy (%)	85.56 ± 1.82	85.19 ± 2.07	90.25 ± 1.42	87.15 ± 1.79
	FPR (%)	2.39 ± 0.18	4.45 ± 0.29	5.22 ± 0.32	7.81 ± 0.49
Over permission	Accuracy (%)	83.13 ± 2.12	86.83 ± 1.79	91.37 ± 1.31	87.93 ± 1.69
	FPR (%)	1.97 ± 0.19	3.51 ± 0.22	4.08 ± 0.18	5.91 ± 0.37

improves detection accuracy because the assessment mechanism becomes more responsive to abnormal behavior accumulation. This trend is particularly evident for SQL injection, buffer overflow, and over permission attacks, where the accuracy increases significantly from $\theta_0 = 0.35$ to $\theta_0 = 0.55$. However, further increasing the threshold does not always lead to better performance.

For example, SQL injection accuracy decreases from 96.82% at $\theta_0 = 0.55$ to 93.89% at $\theta_0 = 0.65$, while the FPR increases from 1.37% to 1.86%. This sensitivity becomes more evident in file inclusion and over permission attacks. These results indicate that excessively high thresholds make the assessment mechanism overly sensitive to borderline behaviors, causing normal activities with slight deviations to be incorrectly classified as abnormal.

Under different threshold settings, DDoS, SQL injection, and buffer overflow attacks achieve higher detection accuracy and lower FPR. This is because DDoS and buffer overflow attacks cause significant degradation in resource availability and system stability, while SQL injection attacks generate distinct abnormal interaction patterns, resulting in clearer characteristic. In contrast, file inclusion and over permission attacks are carried out through legitimate file access or privilege management, causing their behaviors to partially overlap with normal system activities. Such overlap increases classification ambiguity, leading to relatively lower detection accuracy and higher FPR.

Overall, although each attack category may exhibit minor performance fluctuations under different threshold settings, $\theta_0 = 0.55$ achieves the best overall trade-off. This threshold provides sufficient sensitivity for attack detection while maintaining high accuracy and relatively low FPR in the majority of attack scenarios within DMBM.

Fig. 7 illustrates the temporal evolution of PDI under five attack scenarios with different weight configurations. For DDoS and buffer overflow attacks, the PDI values exhibit the most significant initial degradation, indicating severe impacts on system resources and service availability. However, both attacks recover rapidly after defense responses are activated. Under the RMI-oriented configuration ($m_1 = 0.2$, $m_2 = 0.6$, $m_3 = 0.2$), these attacks experience deeper PDI degradation because resource impairment contributes more heavily to the overall assessment. Nevertheless, the recovery levels of DDoS and buffer overflow are improved under this configuration, highlighting the strong correlation between these attacks and resource damage.

In contrast, SQL injection primarily affects the interactions among system components and data dependencies. Therefore, the CRD-oriented configuration ($m_1 = 0.6$, $m_2 = 0.2$, $m_3 = 0.2$) results in a larger initial PDI decline but also provides relatively improved recovery effectiveness. This observation suggests that collaborative relationship damage plays a more important role in characterizing the impact of SQL injection attacks.

File inclusion and over permission attacks exhibit prolonged degradation and slower recovery trends. Although their initial PDI decline is

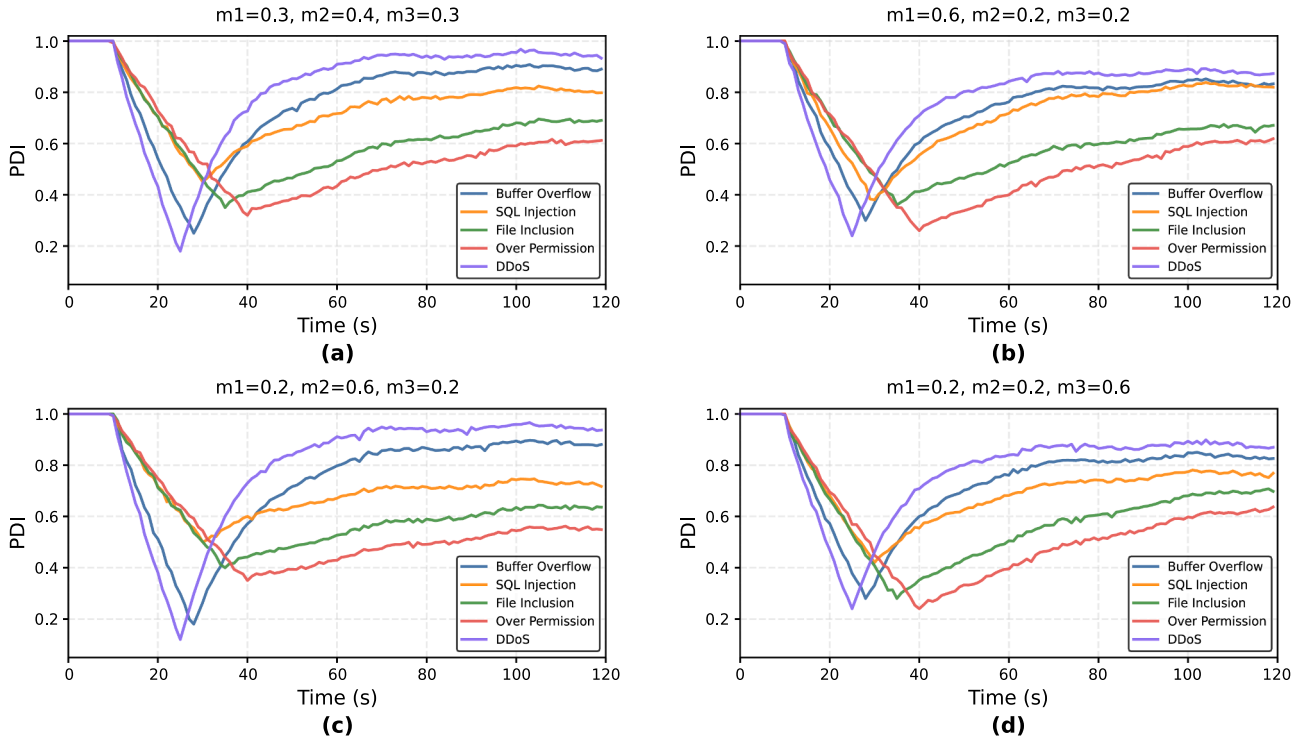


Fig. 7. Temporal evolution of PDI for five attack scenarios under different weight configurations: (a) $m_1 = 0.3$, $m_2 = 0.4$, $m_3 = 0.3$ (DMBM); (b) $m_1 = 0.6$, $m_2 = 0.2$, $m_3 = 0.2$; (c) $m_1 = 0.2$, $m_2 = 0.6$, $m_3 = 0.2$; (d) $m_1 = 0.2$, $m_2 = 0.2$, $m_3 = 0.6$.

not always the most severe, their recovery requires a longer period due to persistent impacts on system structures and operational stability. The result shows that both attacks are affected by CRD-oriented ($m_1 = 0.6$, $m_2 = 0.2$, $m_3 = 0.2$) and SD-oriented ($m_1 = 0.2$, $m_2 = 0.2$, $m_3 = 0.6$) weighting configurations. File inclusion disrupts both object dependencies, while over permission alters control relationships and introduces long-term security risks.

Although specific weighting schemes may improve the recovery performance, none of them consistently outperform the baseline configuration of DMBM ($m_1 = 0.3$, $m_2 = 0.4$, $m_3 = 0.3$). The balanced weighting strategy maintains relatively high recovery levels while avoiding excessive sensitivity to any single indicator. These results demonstrate that the proposed baseline weighting configuration provides a more robust and generalized assessment of process damage and recovery effectiveness.

4.4. Comparison experiment

To further evaluate the effectiveness of our model, four representative approaches, NeuroShield (Tao et al., 2025), SAF-ADM (Li et al., 2023), IIMDA (Adhikari et al., 2025) and FTASH (Prajwalasimha et al., 2025) are selected for comparison. NeuroShield is a bionic-inspired active defense model that incorporates adaptive learning mechanisms to identify malicious behaviors and dynamically update defense strategies. It is capable of improving defense performance over time, but its convergence speed remain limited. In contrast, SAF-ADM adopts defense mechanism by integrating situational awareness with firewall IP blocking. It detects potential attackers based on alarm accumulation, and performs defensive actions through automated blocking. However, its effectiveness heavily relies on threshold settings and lacks adaptive response capability. IIMDA is an immune-inspired malware detection algorithm that leverages antigen-antibody matching and a dynamic memory cell repository to retain historical threat intelligence, enabling rapid recognition of attack patterns, yet its performance remains constrained for threats due to the absence of self-healing mechanism. FTASH is a federated trust-aware and explainable self-healing framework that combines federated learning with Bayesian trust calibration and explainable reinforcement learning, allowing distributed nodes to collaboratively adapt to unknown attacks, but its reliance on communication rounds and consensus aggregation introduces recovery latency and substantial computational overhead.

As shown in Fig. 8, under known attacks, NeuroShield demonstrates a relatively smooth improvement trend due to its adaptive learning

capability, while SAF-ADM improves slowly in the early stages because sufficient alarm accumulation is required before the defense mechanism can be activated. Once the threshold is reached, the detection accuracy increases rapidly but quickly converges to a stable level. IIMDA and FTASH achieve better performance than SAF-ADM by leveraging memory Cell Repository and federated knowledge aggregation mechanisms, respectively. However, their convergence accuracy remains lower than DMBM. Benefiting from dynamic threshold and defense response mechanisms, DMBM achieves the fastest convergence and the highest detection accuracy under known attacks, outperforming the other models by approximately 3%–10%.

For unknown attacks, the performance differences become more pronounced. FTASH achieves the highest accuracy among the contrast models, as its federated learning architecture enables generalization across heterogeneous data distributions, allowing distributed nodes to collectively adapt to emerging threat patterns. IIMDA follows with superior detection accuracy, benefiting from its antigen-antibody similarity matching, which can recognize anomalous patterns even without exact historical precedents. NeuroShield achieves moderate performance, constrained by its reliance on the predefined knowledge base, which limits its perceptual scope against threats falling outside this framework. SAF-ADM performs the poorest, as its threshold-triggered accumulation mechanism lacks the flexibility to respond to attacks that do not generate sufficient alarm signals within the detection window. In contrast, DMBM achieves the highest detection accuracy under unknown attacks, surpassing FTASH, IIMDA, NeuroShield, and SAF-ADM by margins ranging from 2% to 15%. This superiority stems from DMBM's integrated design, which combines the advantages of multimodal perception, self-driven assessment, defense strategy adjustment and immune memory.

4.5. Ablation experiment

The self-healing module and defense strategy are two important mechanisms in this model. To further clarify their impact, we provided a detailed description of the recovery mechanisms and evaluation metrics used in the experiments.

Specifically, the self-healing module performs targeted recovery actions based on the identified damage, including restart, container migration, resource reallocation, and access restriction. For instance, in the case of service disruption caused by DDoS attacks, traffic throttling and resource scaling are applied, while compromised objects are isolated and restarted in buffer overflow and file inclusion scenarios. For

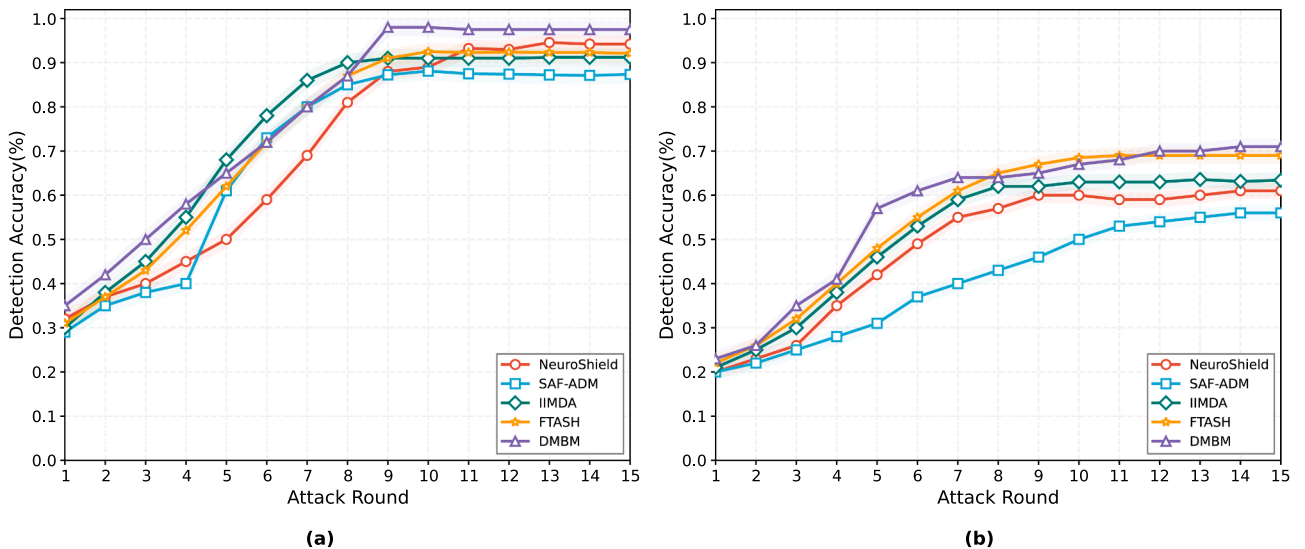


Fig. 8. Detection accuracy of five models under known and unknown attacks: (a) known attacks;(b) unknown attacks.

privilege escalation attacks, abnormal behaviors are terminated and permissions are reset to enforce least-privilege policies. The defense strategy in DMBM is responsible for storing known and unknown attack characteristics, making judgments on the behavior of the target and issuing strategies.

To verify the impact of these two mechanisms on the model's performance, this paper conducted ablation experiments and compared the comprehensive indicators with NeuroShield, SAF-ADM, IIMDA and FTASH, as shown in Table 7.

Table 7 summarizes the performance of different models in terms of TPR, FPR, recovery time, and resource occupation. The proposed DMBM achieves the best overall performance, with the highest TPR (97.45%) and lowest FPR (2.81%), outperforming the contrast models.

The ablation results show that removing the defense strategy leads to a significant performance drop (TPR decreases to 71.38%, FPR increases to 19.93%), indicating that the defense strategy plays a critical role.

From the perspective of damage assessment, the proposed DMBM achieves the lowest PDI value (0.26), indicating the least system damage among all compared models. In contrast, the no-mechanism scenario exhibits the highest PDI (0.71), reflecting severe system degradation without any protection. When the defense strategy module is removed, the PDI decreases to 0.39, suggesting that although basic detection exists, the lack of immune memory leads to insufficient damage control. Similarly, removing the self-healing module results in a higher PDI (0.55), demonstrating that detection alone is not sufficient to mitigate system damage without effective recovery mechanisms.

Compared with other models, NeuroShield (0.41) and SAF-ADM (0.57) both show higher PDI values than DMBM, indicating weaker capability in damage mitigation. Although NeuroShield benefits from adaptive learning, its lack of explicit recovery mechanisms limits its ability to reduce accumulated damage. SAF-ADM performs worse due to its rule-based response strategy, which cannot effectively control damage progression under dynamic attack scenarios. IIMDA achieves competitive detection accuracy through its antigen-antibody matching and memory cell repository, but its PDI remains relatively high because it focuses primarily on malware identification rather than system recovery. FTASH shows the lowest PDI (0.39) among the four contrast models due to its self-healing controller and federated trust calibration, but its PDI still substantially exceeds that of DMBM, suggesting that its distributed consensus mechanism introduces recovery latency and its heavy reliance on global model aggregation delays localized damage recovery.

Moreover, FTASH consumes the highest resource occupation (CPU and memory utilization) among all models, reflecting the substantial computational overhead from its federated learning, Bayesian trust propagation, and XAI modules, which limits its deployment capability in resource-constrained environments.

Although DMBM exhibits the high resource occupation (15.98%), the additional overhead primarily originates from continuous eBPF runtime monitoring and the execution of adaptive self-healing operations. However, this overhead enables DMBM to achieve the highest TPR, the lowest FPR, and the lowest PDI among all compared models, indicating a favorable trade-off between security effectiveness and resource consumption. Nevertheless, the additional overhead may affect deployment efficiency in resource-constrained environments. For typical information systems, where security, resilience, and service continuity are critical requirements, such resource consumption is considered acceptable.

Overall, the results demonstrate that both the defense strategy and the self-healing mechanism are essential for system security. By integrating immune memory and recovery optimization, DMBM effectively suppresses damage propagation and achieves superior resilience.

5. Conclusion

In this paper, we proposed DMBM for information system security.

Table 7

Model performance Comparison.

Model	TPR (%)	FPR (%)	PDI	Resource occupation (%)
No mechanism (DMBM)	70.33	20.41	0.71	5.37
No defense strategy (DMBM)	71.23	20.12	0.39	7.97
No self-healing (DMBM)	96.71	3.32	0.55	5.51
NeuroShield	94.51	5.72	0.41	9.21
SAF-ADM	87.26	4.53	0.57	7.63
IIMDA	91.25	5.79	0.61	10.34
FTASH	92.23	5.10	0.39	21.62
DMBM	97.45	2.81	0.26	15.98

To address the limitations of traditional perimeter-based defense mechanisms, a dynamic defense architecture inspired by biological immune systems is introduced. The key contribution of DMBM lies in unifying dynamic defense, immune memory, and self-healing into a closed-loop security framework. Experimental results demonstrate that DMBM achieves high detection accuracy, low false positive rate, and fast recovery capability. Comparative evaluations with NeuroShield, SAF-ADM, IIMDA and FTASH further verify that the proposed model provides superior performance in both known and unknown attack scenarios. These results indicate that DMBM is effective for real-time security monitoring and resilient defense in complex and dynamic information systems.

Despite its effectiveness, DMBM still has several limitations. On the one hand, the integration of trust evaluation, dynamic detection, and self-healing mechanisms introduces additional computational overhead, which may impact performance in resource-constrained environments.

On the other hand, the performance of the model relies on the quality and completeness of runtime monitoring data (e.g., system calls and behavioral features). In scenarios with incomplete or noisy data, the accuracy of trust evaluation and detection may be affected.

In future work, we plan to further optimize the resource efficiency of DMBM to reduce runtime overhead in large scale deployments. In addition, more advanced learning mechanisms will be introduced to enhance the model's adaptability to unknown attack scenarios. Finally, we aim to extend the model to more complex real-world environments and heterogeneous systems to improve its generalization and practical applicability.

CRedit authorship contribution statement

Yanyi Zhang: Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Tao Li:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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